CO_2 Electroreduction on Cu-Fe Bimetallic Catalyst

Operando Raman Hypothesis Test Report

Catalyst	Cu-Fe 41 (CF41) on screen-printed Au electrode
Electrolyte	CO_2-saturated 0.1 M $KHCO_3$
Experiment Date	2025-07-24 / 2025-10-29
Laser	532 nm (531.95 nm measured)
Conditions	-0.56 V, -0.76 V, -0.80 V, -0.80 V repeat (vs RHE)
Total Scans	45 (10 + 10 + 10 + 15)
Hypothesis	co2rr_cufe_acetate_v1
Analysis Date	2026-03-25

1. Hypothesis Overview

The Cu-Fe bimetallic hypothesis proposes that Fe sites at Cu grain boundaries modify the *CO binding landscape, lowering the barrier for asymmetric C-C coupling (*CO + *CHO) that leads to oxygenate products (acetate, ethanol) rather than symmetric coupling (*CO + *CO) that produces ethylene. The target product is acetate (CH_3COO^-), with the diagnostic Raman peak at 928 cm^{-1} (v(C-C) stretch).

1.1 Reaction Network

CO_2(aq) -> *COOH -> *CO is the central pathway. From *COOH, a branch to formate (C1) competes with further reduction to *CO. From *CO, four branches compete: methanol (C1), ethylene via *CO+*CO (C2, symmetric), ethanol via *CO+*CHO (C2), and the target acetate via Fe-assisted *CO+*CHO (C2). The rate-limiting step is C-C coupling (R5, tau ~ 120 s on Cu).

1.2 Predictions by Potential

Potential	Regime	Dominant products	Acetate prediction	Key test
-0.56 V	Low overpotential	Formate, *CO	Absent (neg. control)	If acetate appears, Fe effect stronger than predicted
-0.76 V	Moderate (critical)	Formate, *CO, acetate	Should emerge (>90 s)	Acetate/ethylene ratio is key diagnostic for Fe promotion
-0.80 V	High overpotential	Acetate, ethylene, formate	Major C2 (>60 s)	Acetate/ethylene >1.0 on Cu-Fe vs <0.1 on pure Cu

2. Scan-by-Scan Peak Analysis

2.1 Condition: -0.56 V vs RHE (10 scans)

Scan	Time	Hypothesis species	Electrode	Electrolyte	Unassigned
1	20:42	Ethylene 1637*	CuO 639	-	2172, 3228, 3423
3	20:47	Ethylene 1635*	-	H_2O 3408	2023, 3235
5	20:51	*CO 2009 (low)	Cu_2O 615, CuO 637	H_2O 1642, 3402	-
7	20:55	None	Cu_2O 615	H_2O 1648, 3404	-
10	21:01	None	-	H_2O 1638, 3404	3478

* "Ethylene" at 1635-1637 cm^{-1} is almost certainly the water bending mode (ref 1640 cm^{-1}). True ethylene ν(C=C) is 1623 cm^{-1} - the 12-14 cm^{-1} shift is too large. Key observation: CuO -> Cu_2O reduction sequence across scans 1-7.

2.2 Condition: -0.76 V vs RHE (10 scans)

Scan	Time	Hypothesis species	Electrode	Electrolyte	Unassigned
1	19:24	None	-	H_2O 3414	1670, 1883
5	19:28	CO_2(aq) 1386	-	H_2O 3393	484, 2901
8	19:31	CO_2(aq) 1386	Fe_2O_3 486	H_2O 3404	2327
10	19:33	CO_2(aq) 1285+1386	-	H_2O 3405	484, 1675, 3216

Key observations: (1) Both CO_2 Fermi dyad members detected - CO_2 is not being consumed. (2) Fe_2O_3 at ~484 cm^{-1} indicates Fe oxidation under operating conditions. (3) No CO_2RR products or intermediates detected.

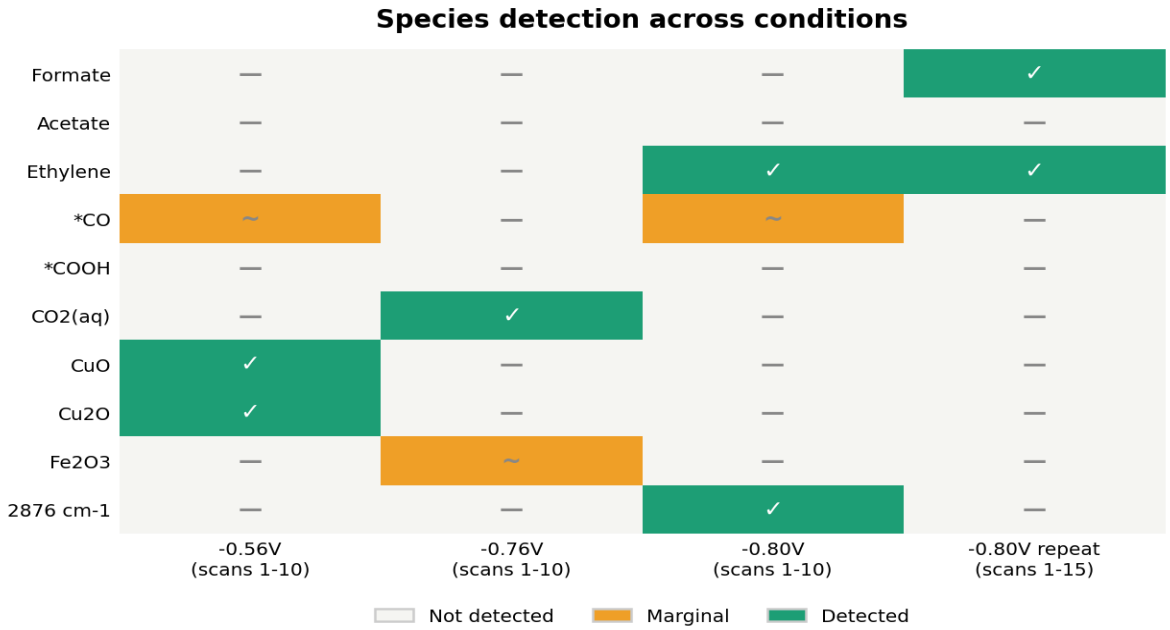
2.3 Condition: -0.80 V vs RHE (10 scans + 15 repeat)

Scan	Run	Hypothesis species	Electrode	Electrolyte	Unassigned
1	First	None	-	-	1666, 2876, 3208, 3422
5	First	None	-	H_2O 1653	2876, 3230, 3383
10	First	Ethylene 1629	-	H_2O 3403	2029, 2876, 3205
1	Repeat	None	-	H_2O 1653, 3393	2329, 3210
8	Repeat	Ethylene 1614	-	H_2O 3406	473, 2329, 3197
15	Repeat	Ethylene 1635, Formate 1342	-	H_2O 3406	2332, 3200

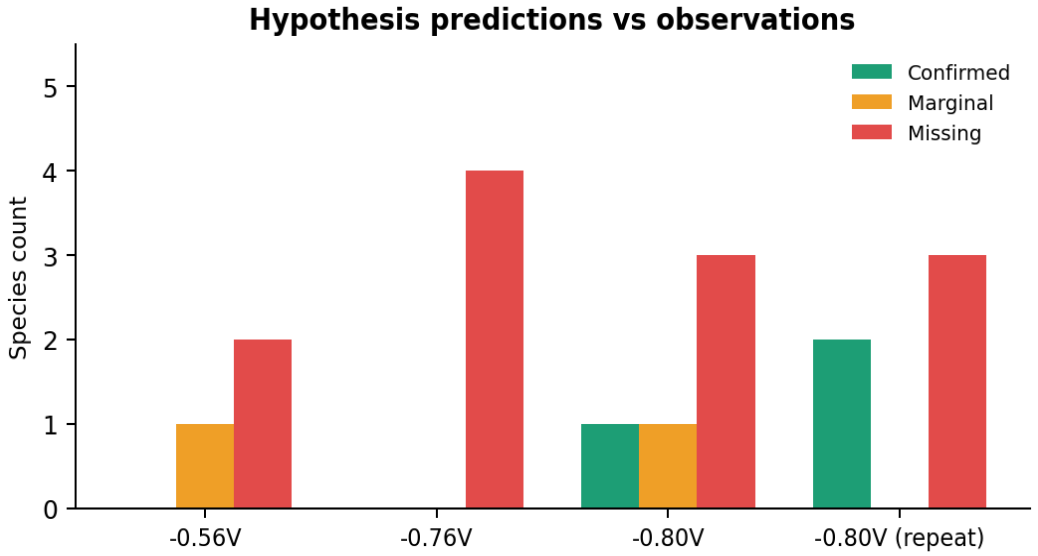
Ethylene confirmed at -0.80 V (both runs). Formate appears only at scan 15 of repeat. Persistent 2876 cm^{-1} feature at -0.80 V only. Acetate 928 cm^{-1} region is flat baseline.

3. Visual Summary

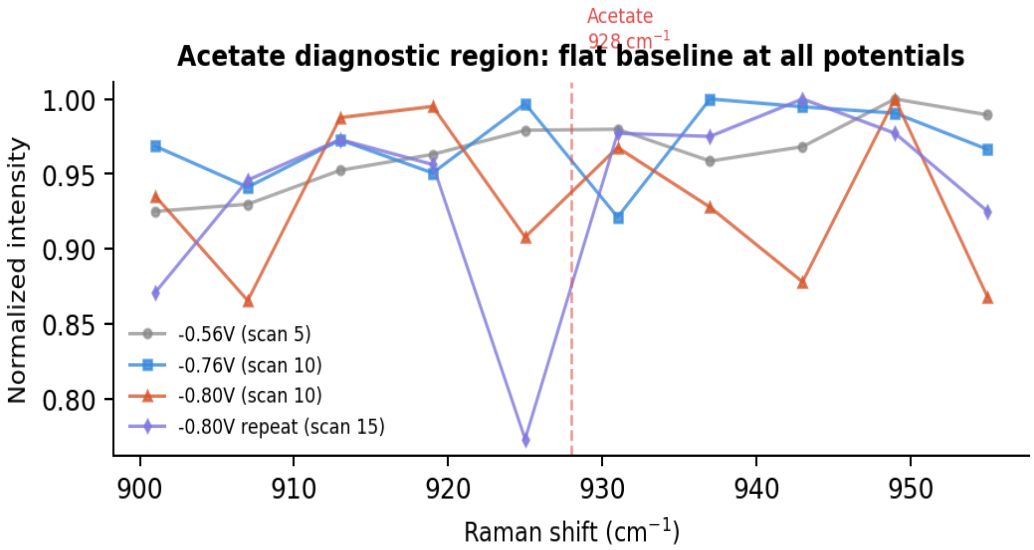
3.1 Species detection heatmap



3.2 Hypothesis score by condition



3.3 Acetate diagnostic region (928 cm^{-1})


Acetate diagnostic region: flat baseline at all potentials
Acetate
928 cm⁻¹
Normalized intensity
1.00
0.95
0.90
0.85
0.80
900
910
920
930
940
950
Raman shift (cm⁻¹)
-0.56V (scan 5)
-0.76V (scan 10)
-0.80V (scan 10)
-0.80V repeat (scan 15)

4. Hypothesis Scorecard

Species	-0.56 V	-0.76 V	-0.80 V	Hypothesis role
Acetate (928)	Absent ✓	Missing ✗	Missing ✗	TARGET: Fe-assisted C-C coupling
Ethylene (1623)	Absent ✓	Missing ✗	Detected ✓	Secondary C2 at high overpotential
Formate (1351)	Missing ✗	Missing ✗	Late only	Dominant C1 at low overpotential
***CO (1990-2060)**	Marginal	Missing ✗	Marginal	Central intermediate
***COOH (1540)**	Missing ✗	Missing ✗	Missing ✗	Early transient intermediate
CO_2(aq) (1285/1388)	-	Unexpected	-	Not tracked in hypothesis
CuO / Cu_2O	Unexpected	-	-	Not in model
Fe_2O_3 (~484)	-	Unexpected	-	Not in model - potentially critical

4.1 Unassigned peaks requiring attention

Wavenumber	Condition	Persistence	Nearest ref	Possible assignment
2876	-0.80 V (all)	Every scan, SNR 5.5-6.5	MeOH 2833 (+43)	Generic v(C-H); hydrocarbon deposit
~3200	-0.80 V (all)	Every scan, SNR 3.8-5.8	H₂O (-200)	Structured O-H; interfacial water
2172	-0.56 V (scan 1)	Single scan	*CO 2060 (+112)	Surface CN^- or isocyanate
~2329	-0.80 V repeat	Persistent	-	Atmospheric N_2 Q-branch (2331)
484	-0.76 V	Scans 5-10	Fe_2O_3 500 (-16)	Iron oxide - Fe speciation issue

5. Verdicts and Interpretation

Verdict 1: Acetate is absent at all potentials - the central hypothesis is not supported

The 928 cm^{-1} diagnostic region is flat baseline across all conditions including -0.80 V. The hypothesis predicted acetate/ethylene >1.0 at -0.80 V; the observed ratio is 0. Fe sites do not appear to redirect C-C coupling toward acetate under these conditions.

Verdict 2: Ethylene - not acetate - is the only C2 product, and only at -0.80 V

Ethylene (1614-1635 cm^{-1}) emerges in late scans at -0.80 V, confirming C-C coupling occurs at high overpotential via symmetric *CO dimerization. The Cu-Fe catalyst behaves like standard Cu for C-C coupling selectivity.

Verdict 3: Intermediates are far weaker than predicted

*CO was marginal at -0.56 V and -0.80 V, absent at -0.76 V. *COOH was never detected at any condition. This suggests either very low reaction rates, poor SERS enhancement on this particular Cu-Fe surface, or both.

Verdict 4: The most informative signals were unexpected

Dissolved CO_2 (both Fermi dyad members) persists at -0.76 V, confirming low consumption rate. Fe_2O_3 at ~484 cm^{-1} indicates Fe sites are oxidized under operating conditions - this may explain why Fe promotion fails. CuO -> Cu_2O reduction at -0.56 V tracks oxide-derived Cu formation affecting SERS enhancement.

Verdict 5: Persistent 2876 cm^{-1} feature at -0.80 V needs investigation

Present in every -0.80 V scan (SNR 5.5-6.5) but absent at milder potentials. In the v(C-H) region; candidates include surface hydrocarbon deposit, adventitious carbon, or an unidentified C1/C2 product.

6. Mechanistic Explanations

Three possible explanations for why the Fe promotion effect is not observed:

6.1 Fe_2O_3 poisoning (most specific)

Fe_2O_3 detected at ~484 cm^{-1} at -0.76 V. If the Fe component oxidizes to hematite under CO_2RR conditions rather than remaining as metallic Fe or FeO_x at Cu grain boundaries, it cannot serve as a *CO binding modifier. The Fe_2O_3 A_{1g} mode is diagnostic of fully oxidized Fe(III), not the reduced Fe(0) or Fe(2+) the hypothesis requires. This would mean the hypothesis mechanism may be correct in principle but the catalyst preparation does not deliver the correct Fe speciation.

*6.2 Insufficient *CO coverage*

The hypothesis assumes high *CO surface coverage as a prerequisite for C-C coupling. The marginal *CO signals across all conditions suggest coverage never reaches the threshold. At -0.56 V the reaction rate is too low; at -0.76 V Fe_2O_3 may block Cu active sites; at -0.80 V, *CO may form and dimerize quickly via the standard Cu pathway without sufficient residence time for asymmetric coupling.

6.3 SERS enhancement mismatch

The dramatic signal decay at -0.56 V as copper oxides reduce, and the much lower overall intensities at -0.76 V and -0.80 V, suggest a sensitivity problem. The oxide-derived Cu roughening that enhances SERS at early scans is lost as the surface smooths. Solution-phase products like acetate lack the surface-enhancement factor and may require much higher concentrations for Raman detection.

7. Proposed Hypothesis Revisions

[ADD] Electrode surface evolution module

Model the CuO -> Cu_2O -> Cu(0) reduction sequence as a function of potential and time. This controls SERS enhancement factor and active site density.

[ADD] Fe speciation as a gating variable

Add Fe_2O_3 as a possible electrode species. The hypothesis should predict that Fe promotion only works if Fe remains as Fe(0) or FeO_x - not as bulk Fe_2O_3. The 484 cm^{-1} mode becomes a negative indicator.

[MODIFY] Lower expected intermediate intensities

Revise *CO and *COOH expected relative concentrations downward by 5-10x, or add a SERS correction factor acknowledging that surface intermediates may be invisible on smooth metallic Cu.

[MODIFY] Shift formate onset prediction

Formate was predicted as an early product (<60 s). Observed: only at scan 15 of -0.80 V repeat. Either the onset potential is higher than assumed, or solution-phase formate requires longer accumulation.

[DEMOTE] R7: Fe-assisted acetate pathway

Zero acetate signal at any condition. Demote from "target pathway" to "unconfirmed hypothesis - requires Fe speciation control." Add explicit precondition: only active if Fe_2O_3 is absent.

8. Recommended Next Experiments

#	Priority	Experiment	Rationale
E1	Critical	**XPS or ex situ Raman on fresh vs. used Cu-Fe 41**	Determine Fe oxidation state before and after electrolysis. If Fe is Fe_2O_3 even before CO_2RR, catalyst needs re-preparation with reduction pretreatment.
E2	Critical	**Pre-reduction protocol at -1.0 V in Ar-purged electrolyte** (5-10 min before CO_2)	Fully reduce Cu and Fe oxides, ensuring metallic Fe is available. Then repeat Raman at -0.76 V and -0.80 V.
E3	Important	**Pure Cu control electrode under identical conditions**	If ethylene signal and formate timing are identical on pure Cu, Fe in Cu-Fe 41 is a spectator or poison.
E4	Important	**Extended electrolysis at -0.80 V (30+ scans)**	Track 2876 cm^{-1} feature: monotonic growth = carbon deposition; stabilization = steady-state product.
E5	Informative	**Offline product analysis (GC or HPLC) of electrolyte after -0.80 V**	Independently confirm whether acetate is truly absent or just below Raman detection limits.
E6	Informative	**Monitor 484 cm^{-1} region at -0.80 V**	If Fe_2O_3 is absent at -0.80 V (more negative potential reduces Fe), supports Fe speciation gating model.

9. Summary

The Cu-Fe bimetallic acetate hypothesis (co2rr_cufe_acetate_v1) is **not supported** by the operando Raman data from the 2025-07-24 and 2025-10-29 sessions. Acetate was not detected at any potential. The catalyst behaves like standard Cu for C-C coupling selectivity, producing ethylene only at -0.80 V. However, the detection of Fe_2O_3 at -0.76 V provides a specific, testable rescue mechanism: Fe oxidation under operating conditions may prevent the promotion effect from functioning. The critical next step is to determine Fe speciation (E1) and test a pre-reduction protocol (E2). If acetate still does not appear after ensuring metallic Fe is present, the hypothesis requires fundamental revision.

[redacted]